

Effects of acute stress on executive functions in depression, generalised anxiety and borderline personality disorder

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ABSTRACT

Background: Acute stress adaptively alters executive functions (EFs) essential for emotion regulation. Emerging systems of psychiatric classification, such as the Hierarchical Taxonomy of Psychopathology (HiTOP) and the Research Domain Criteria, emphasise the underlying mechanisms and dimensional nature of psychopathology. Distress disorders—a subfactor within the HiTOP model including major depressive disorder, generalised anxiety disorder and borderline personality disorder—are characterised by altered stress reactivity, impaired emotion regulation, and modest responsiveness to first-line psychotherapies. This systematic review sought to examine whether distress disorders and their related symptoms confer heightened vulnerability to transient EF impairment under conditions of acute stress.

Methods: A comprehensive search of articles published in ProQuest, PsycINFO, PubMed, Scopus and Web of Science prior to December 31st, 2022 identified 17 suitable studies examining stress-induced alterations to working memory, inhibition and cognitive flexibility in the context of distress disorders and associated symptoms.

Results: This review found a heightened susceptibility to stress-induced impairment of working memory in depression, and of response inhibition in borderline personality disorder—even for sub-clinical presentations of depressive symptoms wherein diagnostic criteria for a depressive disorder were not met. Findings for cognitive flexibility were inconclusive.

Limitations: While a thorough systematic review was conducted, heterogeneity in study design and methodologies precluded inclusion of a meta-analysis.

Conclusions: The findings indicate that altered stress reactivity leads to maladaptation of EF under acute stress, and subsequent disruption of adaptive emotion regulation strategies. The present finding may help account for non-responsiveness among therapeutic interventions predicated upon cognitively-demanding emotion regulation strategies.

1. Introduction

Exposure to acute stress is associated with both improvements and impairments in executive function (EF) processes such as working memory (WM), inhibition and cognitive flexibility (Shields et al., 2016). EFs are crucial to emotion regulation, underpinning, for instance, the shifting of attention from an unpleasant stimulus, or the reappraisal of a pessimistic first impression (Ochsner et al., 2012; Toh and Yang, 2022a, 2022b). In this review, we attempt to clarify the effects of acute stress on EF in distress disorders, a set of conditions characterised by elevated

stress reactivity, impaired emotion regulation, and modest responsiveness to first-line psychological interventions (Borkovec and Ruscio, 2001; Mennin and Fresco, 2013; Renna et al., 2017; Zorn et al., 2017; Drews et al., 2019; Im, 2021). Distress disorders include generalised anxiety disorder (GAD), major depressive disorder (MDD), dysthymic disorder, borderline personality disorder (BPD), and posttraumatic stress disorder (PTSD; Kotov et al., 2021). Given the modest responsiveness of these disorders to first-line therapies, we aimed to explore a possible role of transient, stress-linked executive impairment in both the pathogenesis and maintenance of these conditions.

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This systematic review attempts to synthesise the literature on the effects of acute stress on executive functions in distress disorders, namely MDD, GAD and BPD. In keeping with an emerging dimensional approach to psychiatric classification, we included studies examining depression and anxiety symptoms among non-clinical participants. EF impairments have been implicated in distress psychopathology (e.g., [Gotlib and Joormann, 2010](#)). However, the influence of transient, situational stressors on cognition is less well understood. We consider that heightened stress reactivity, such as in distress disorders, may confer greater susceptibility to transient EF disruptions under acute stress. Such disruptions would carry clear implications for therapeutic interventions predicated on the use of cognitively-demanding emotion regulation strategies in emotionally-stressful contexts. We therefore aim to build upon existing meta-analytic evidence for stress-linked EF alterations among non-clinical populations ([Shields et al., 2016](#)). First, we discuss the effect of stress on prefrontal executive control regions, the distress disorder construct, alterations in physiological stress reactivity in distress disorders, and contemporary models of EF. We then discuss meta-analytic evidence of acute stress effects on executive functions in non-clinical populations. To address the primary aim of the review, we then present a systematic review of empirical studies examining stress-induced EF alterations in the context of distress disorders and associated symptoms. Finally, we discuss theoretical and clinical implications of our findings, discuss transient EF disruptions as a possible contributing factor to the modest efficacy of therapeutic treatments, and propose directions for further investigation.

1.1. Stress and prefrontal functioning

Acute stress has been proposed to alter cognition in a manner which promotes rapid behavioural responses to emergent stressors ([Hermans et al., 2014](#); [Thayer and Lane, 2000](#); [Vogel et al., 2016](#)). Acute stress leads to the disruption of executive processes including WM ([Arnsten, 2009](#); [Shields et al., 2016](#)), cognitive flexibility ([Alexander et al., 2007](#); [Gabrys et al., 2019](#)) and inhibition ([Alomari et al., 2015](#); [de Berker et al., 2016](#)). A putative mechanism of this adaptation is rapid impairment of prefrontal functioning under stress, leading to a reversal of top-down prefrontal inhibition of subcortical regions ([Arnsten, 2009](#)). The prefrontal cortex (PFC) is a primary neural substrate of EFs ([Diamond, 2013](#)). Under non-stressful conditions, the PFC exerts inhibitory control over subcortical regions associated with emotion processing, such as the amygdala ([McKlveen et al., 2015](#)). However, under heightened stress the amygdala inhibits regions of the PFC, impairing prefrontal executive processes such as WM and disinhibiting reflexive actions ([Arnsten, 2009](#); [Liston et al., 2009](#); [Thayer et al., 2009](#)). This adaptation is crucial in contexts where urgent action is preferable to cognitive engagement. For example, escape from an immediate threat may be better served by fast, automatic and habitual action than by slower goal-directed reasoning ([Arnsten, 2009](#); [Felix-Ortiz et al., 2016](#); [Thayer and Lane, 2000](#); [Evans and Stanovich, 2013](#); [Yu, 2016](#)).

1.2. Executive functions and distress disorders

Adaptation of EFs under stress carry implications for emotional disorders associated with altered stress reactivity. An emerging trans-diagnostic approach to psychopathology proposes that depressive disorders (including MDD and dysthymia), GAD, PTSD and BPD can be grouped under a 'distress disorders' subfactor, based on substantial comorbidities ([Kotov et al., 2017](#); [Krueger et al., 2018](#)). Etiologically, distress disorders are proposed to share a common basis of altered stress reactivity ([Clark and Watson, 2006](#); [Kotov et al., 2021](#)). Cognitive behavioural therapy (CBT) and derivatives such as acceptance and commitment therapy and dialectical behaviour therapy are widely considered as first-line psychotherapeutic treatments for these disorders ([Andrews et al., 2018](#); [Gautam et al., 2017](#); [Kendrick et al., 2022](#); [Qaseem et al., 2023](#); [Sharan et al., 2023](#); [Stone, 2019](#)). These therapeutic

approaches are largely predicated on the use of problem-focused strategies to improve emotion regulation, such as attentional training (i.e., mindfulness), distraction and reappraisal ([Arch and Craske, 2009](#); [Fassbinder et al., 2016](#); [Gross, 2015](#)). However, converging evidence suggests relatively modest responsiveness to these therapies among distress disorders, despite statistically impressive results in randomised controlled trials. For instance, despite large effect sizes pertaining to cognitive behavioural therapy for GAD ([Carpenter et al., 2018](#)), as few as 50 % of patients may achieve clinically significant improvement ([Borkovec and Ruscio, 2001](#); [Renna et al., 2017](#)). A comparative meta-analysis of BPD treatments found that modest improvements reported for dialectical behaviour therapy could not be reliably demonstrated at follow-up, and that CBT yielded no benefit over control conditions ([Cristea et al., 2017](#)). Similarly, Cuijpers and colleagues (2013) found no support for the superiority of CBT over other therapies in the treatment of MDD. Drop-out rates among these therapies tend to be high ([Dixon and Linardon, 2020](#); [Fernandez et al., 2015](#); [Ong et al., 2018](#)), suggesting that many patients do not find these therapies beneficial. Comorbid presentations of these disorders, which are highly prevalent ([Kotov et al., 2017](#)), may be particularly refractory to treatment ([Farabaugh, 2012](#)). Therefore, clarification of factors contributing to treatment non-response is needed to better inform clinical guidelines.

One possible factor concerns the reliance of emotion regulation strategies such as mindfulness, distraction and reappraisal, on EFs such as working memory, inhibition and cognitive flexibility ([Buhle et al., 2014](#); [Ochsner et al., 2012](#); [Toh and Yang, 2022a, 2022b](#); [Wager et al., 2008](#)). Heightened stress impairs EFs ([Alexander et al., 2007](#); [Arnsten, 2009](#); [Alomari et al., 2015](#); [de Berker et al., 2016](#); [Shields et al., 2016](#); [Gabrys et al., 2019](#)). However, moderate stress is thought to broadly enhance adaptive functioning ([Frankenhuis and de Weerth, 2013](#); [Calabrese and Mattson, 2017](#); [Crane et al., 2019](#); [Höltge et al., 2019](#)). Such effects may therefore follow an inverted-U function, with moderate stress leading to EF improvement and severe stress to EF impairment ([Yuen et al., 2009](#); [Sapolsky, 2015](#); [Oshri et al., 2022](#)). This, in turn, implies that the success of emotion regulation strategies may depend in part on whether acute stress imparts a beneficial or deleterious effect upon the EFs which underpin them. Indeed, findings are mixed regarding the effect of acute stress on distraction and reappraisal among non-clinical participants, with some studies reporting stress-linked improvements in successful emotion regulation ([Kinner et al., 2014](#); [Langer et al., 2021, 2022](#); [Langer et al., 2023](#)) and others reporting stress-linked impairments ([Raio et al., 2013](#); [Zhan et al., 2017](#)). The altered stress reactivity observed in distress disorders may therefore confer susceptibility to EF impairment under stress. Consequent impairment of emotion regulation strategies may help explain the modest efficacy of first-line therapies in these disorders—a suggestion borne out by observations of impaired emotion regulation among these conditions ([Johnstone et al., 2007](#); [Mennin and Fresco, 2013](#); [Renna et al., 2017](#); [De la Peña-Arteaga et al., 2021](#); [Keller, 2022](#)).

1.3. Dimensional approaches to classification

Conventional diagnostic systems imperfectly reflect the nature of mental disorders ([Kotov et al., 2017, 2021](#)). Both the *Diagnostic and Statistical Manual of Mental Disorders* ([American Psychiatric Association, 2022](#)) and the *International Classification of Diseases* ([World Health Organization, 2019](#)) specify diagnostic categories derived largely from expert consensus. Many categories demonstrate poor inter-rater and test-retest reliability ([Chmielewski et al., 2015](#); [Regier et al., 2013](#)), excessive within-diagnosis heterogeneity ([Galatzer-Levy and Bryant, 2013](#)) and excessive comorbidities ([Clark et al., 2017](#); [Newman et al., 2010](#); [Newson et al., 2021](#)). A growing recognition of such limitations has prompted efforts to reconceptualise psychiatric classification to reflect the dimensional nature of mental disorders, to better elucidate their etiological bases, and to more effectively discriminate between etiologically-distinct conditions. The Research Domain Criteria (RDoC)

and the Hierarchical Taxonomy of Psychopathology (HiTOP) represent two major frameworks emerging from this dimensional perspective. The RDoC, an initiative of the National Institute of Mental Health, delineates various domains of psychological constructs (e.g., cognitive systems, social processes), and specifies cross-cutting units of analysis (e.g., genetic, behavioural). This conceptual matrix guides research into the basic systems and processes implicated in psychiatric disorders, representing a focus on the mechanisms of psychological dysfunction spanning relatively normal, subclinical and clinical presentations (Cuthbert and Insel, 2013; Cuthbert, 2022).

The Hierarchical Taxonomy of Psychopathology (HiTOP) model represents an alternative taxonomy derived from latent-variable analysis of dimensional measures of symptoms, syndromes, and personality traits (Kotov et al., 2017, 2021). In contrast to the emphasis in the RDoC framework on the mechanisms of psychological dysfunction, the HiTOP framework focuses on the co-occurrence of clinical phenomena (Micheline et al., 2021). Within the HiTOP architecture, conditions corresponding to DSM-5 categories of MDD, dysthymia, GAD, BPD and PTSD are subsumed under the ‘distress disorders’ subfactor based on high comorbidities between these conditions (Kotov et al., 2017). Conceptions of the distress subfactor predated the release of the DSM-5 in 2013, with MDD and GAD found to load onto a latent factor thought to be underpinned by neuroticism or negative temperament (Clark and Watson, 2006; Watson et al., 2008). Distress disorders appear to be distinct from a closely related ‘fear disorders’ subfactor comprising panic disorder, social anxiety disorder and specific phobia. Whereas fear disorders are characterised by heightened physiological reactivity and fears of physiological arousal, MDD and GAD are characterised by blunted physiological reactivity and fears of cognitive arousal, such as loss of control over one’s thoughts (Allan et al., 2015; Hyde et al., 2019; Saulnier et al., 2018).

1.3.1. Altered stress reactivity in distress disorders

Exposure to stressful stimuli elicits a cascade of physiological responses, which are organised to support behavioural responses to threat—the so-called ‘fight-or-flight’ response (Kozłowska et al., 2015). These responses include, among others, upregulation of stress hormones such as cortisol, and modulation of sympathetic and parasympathetic tone within the autonomic nervous system (Kozłowska et al., 2015). Converging evidence from psychophysiology (including neuroimaging) and cortisol assay studies points to altered stress reactivity in distress disorders. Blunted parasympathetic responsiveness to induced stress has been reported among patients with MDD (Bylsma et al., 2014; Nugent et al., 2011; Rottenberg et al., 2007), GAD (Im, 2021; Levine et al., 2016) and BPD (Lozovanu et al., 2020; Weinberg et al., 2009). A comprehensive meta-analysis (Zorn et al., 2017) found that among MDD patients, cortisol reactivity was blunted among women and enhanced among men. These differences were suggested to relate, in part, to the differential influences of sex hormones (Foley and Kirschbaum, 2010; Juster et al., 2016). A recent review reported a general pattern of elevated basal cortisol, aberrant cortisol reactivity and delayed returns to basal cortisol levels among MDD patients, following acute stress induction (Nandam et al., 2020). A meta-analysis of stress-induction studies on participants with BPD revealed a similar pattern of elevated basal cortisol, blunted cortisol response and impaired recovery, though no gender differences were identified (Drews et al., 2019). Regarding neuroimaging evidence, prefrontal hypoactivation in response to stress has been reported among MDD (Keller et al., 2022), GAD (Ball et al., 2013) and BPD (Aguilar-Ortiz et al., 2020) patients, and impaired PFC-subcortical coupling under stress was recently reported in MDD (Hosseini et al., 2022).

Taken together, these alterations may reflect a generalised pattern of heightened physiological arousal in distress disorders, possibly contributing to their high co-morbidities (Kotov et al., 2021). For instance, blunted cortisol reactivity may reflect an elevated basal cortisol set-point (Nandam et al., 2020), and blunted parasympathetic responsiveness may reflect autonomic arousal (Thayer and Lane, 2000).

Furthermore, stress-induced loss of prefrontal functioning may undermine executive resources vital to the recruitment of emotion regulation strategies, such as reappraisal (Toh and Yang, 2022a, 2022b). It follows, therefore, that generalised physiological arousal in distress disorders may be compounded by compromised emotion regulation, due to liability of the PFC to rapid deactivation under emotional stress.

1.4. Executive functions

Executive functions (EF) refer to a family of top-down mental processes involved in the coordination of cognitive control over behaviour (Diamond, 2020). EF processes are effortful, in that the recruitment of cognitive control over automatic, reflexive behaviour involves the inhibition of prepotent, stimulus-driven responses, volitional direction of attention, active interference control, and the maintenance, manipulation and updating of information (Chai et al., 2018; Diamond, 2013, 2020).

The dominant contemporary account of EF (Diamond, 2020) originated from latent-variable analyses by Miyake and colleagues (2000). These analyses suggested dissociable but interrelated cognitive operations related to the updating of WM, the inhibition of prepotent responses and the flexible shifting of mental operations between tasks. This overall structure was initially replicated among various samples, however was subsequently amended to reflect specific updating and shifting functions alongside a common executive factor (Friedman et al., 2008, 2016). Researchers have begun to explicate the neural substrates of these functions throughout the PFC and associated networks (Friedman and Miyake, 2017). Recent developments indicate a two-factor model wherein ‘cool’ EFs (WM, inhibition and cognitive flexibility) govern how information is processed, and ‘hot’ EFs (e.g., reward processing, risky decision-making, delay discounting) regulate emotions and motivation in emotionally-significant situations (Fiske and Holmboe, 2019; Montroy et al., 2019; Salehinejad et al., 2021). This distinction is reflected in structural differences, with cool EFs associated with PFC activity and hot EFs associated with activity in subcortical structures such as the amygdala and insula, as well as select prefrontal regions with robust subcortical projections (i.e., medial PFC; Salehinejad et al., 2021).

1.4.1 Executive Functions and Acute Stress. A substantial evidence base suggests that exposure to acute stress leads to EF alterations. A common stressor in daily life is social-evaluative threat, the perceived threat of negative appraisal (Dickerson and Kemeny, 2004). Social-evaluative stress readily elicits heightened subjective stress and pronounced responses in the hypothalamic-pituitary-adrenal axis (HPA), autonomic nervous system (ANS) and other physiological systems. Experimental manipulation of acute stress is often achieved through validated stress induction protocols, such as the Trier Social Stress Test (TSST; Kirschbaum et al., 1993). In the TSST, participants deliver an unscripted speech and complete a verbal arithmetic task, while social-evaluative threat is purportedly elicited through observation by an unresponsive audience. The TSST reliably elicits strong subjective and physiological stress responses, as indexed with self-report measures, indices of HPA arousal such as salivary cortisol, and cardiac indices of ANS response (Kirschbaum et al., 1993). Other common methods of acute stress induction include the pairing of cognitive tasks with aversive stimuli, as in the Mannheim Multicomponent Stress Test (MMST; Kolotylova et al., 2010), and administration of pharmacological agents to directly elicit physiological stress responses (e.g., Carvalho Fernando et al., 2013). Such methods offer ecological validity in approximating common sources of stress in daily life, such appraisal of one’s performance, aversive distractors during a difficult task, or hormonal responses to stressful stimuli.

In the most comprehensive review to date, Shields et al. (2016) performed a meta-analysis of studies examining the effects of acute stress induction on WM (34 studies), inhibition (21 studies) and cognitive flexibility (6 studies) among healthy participants. The reviewers

found that acute stress impaired performance on tasks of WM, cognitive flexibility and cognitive (i.e., attentional) inhibition, but improved performance on tasks of response (i.e., behavioural) inhibition (for a review of these components of inhibition, see [Tiego et al., 2018](#)). This pattern of results was suggested to imply an adaptation of executive functions under stress, wherein EFs supporting executive control over cognition are impaired in favour of enhanced motor control (improved response inhibition). This adaptation was argued to correspond to a ‘fight-or-flight’ state ([Kozłowska et al., 2015](#)), briefly narrowing one’s cognitive abilities to force attention toward salient, threat-related information, while at the same time enhancing one’s control over motor responses relevant to, for instance, countering or evading a threat. However, a relative paucity of studies on cognitive flexibility precluded strong conclusions. The reviewers also found that effect sizes were substantially moderated by the degree of experimental control over confounds such as medication and hormonal contraceptives, indicating potential quality concerns among the included studies. Notwithstanding these limitations, the observed adaptation of EFs under stress implies that heightened stress reactivity—as observed in distress disorders—may lead to differential alteration of EFs relative to healthy participants.

1.5. The current study

[Shields et al. \(2016\)](#) findings suggest that acute stress leads to an adaptation of executive functions, that constrain certain ‘cool’ EF capacities (WM, attentional inhibition and cognitive flexibility) to direct information processing toward salient stimuli, while enhancing control over motor responses. One possible mechanism for this adaptation is the reversal of prefrontal inhibition of subcortical structures under stress, resulting in diffuse PFC deactivation alongside the typical physiological stress response ([Kozłowska et al., 2015](#)). This adaptation may be relevant to distress disorders: To review, distress disorders are associated with excessive prefrontal deactivation and blunted physiological reactivity, possibly reflecting a generalised physiological arousal. The burden of additional executive impairment under stress may help to explain findings of modest treatment response to cognitive therapies among individuals with distress disorders ([Borkovec and Ruscio, 2001](#); [Farabaugh, 2012](#); [Cristea et al., 2017](#); [Newman et al., 2010](#); [Woodbridge et al., 2021](#)).

This paper presents a synthesis of studies that have investigated acute stress-induced EF alterations in the context of distress disorders and related symptoms. Whereas prior reviews have examined EF deficits in specific disorders, such as inhibition impairments in BPD ([McHugh and Balaratnasingam, 2017](#); [Turner et al., 2017](#)), the current study represents an attempt to systematically investigate stress-related EF adaptations as they pertain to a subset of distress disorder (MDD, GAD and BPD). We set out to examine whether these disorders confer a heightened vulnerability to the stress-induced impairment of working memory, cognitive inhibition and cognitive flexibility observed by [Shields et al. \(2016\)](#). In keeping with a dimensional approach to psychiatric diagnosis, studies featuring sub-clinical presentations of depression and anxiety symptoms were included for review alongside those with participants meeting DSM or ICD criteria for MDD, GAD or BPD. The method and results are reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (see Appendix A for the PRISMA checklist).

2. Method

2.1. Scoping search for meta-analytic review

A pilot search of the literature was undertaken to assess the feasibility of a meta-analytic review in addressing the study aims. For continuity with existing research, it was planned to replicate [Shields et al. \(2016\)](#) meta-analysis. We further planned to extend their findings

through the inclusion of studies with MDD, GAD and BPD participants, and the analysis of disorder status as a moderator between acute stress induction and executive functions. Peer-reviewed studies were identified via English-language keyword searches, using [Shields et al. \(2016\)](#) original search terms, conducted on the Web of Science, Scopus, PsycINFO, ProQuest and PubMed databases, for papers published between January 1st, 1900 and December 31st, 2022. These searches yielded 682 results from Web of Science, 554 results from Scopus, 428 results from PsycINFO, 186 results from ProQuest and 1326 results from PubMed. After removal of duplicates, 1855 articles remained. Abstracts were dual screened for inclusion based on the following criteria: (1) experimental design; (2) induction of acute stress in human participants ≥ 18 years of age; (3) subsequent completion of a task established to index WM, cognitive flexibility or inhibition (not more than one-hour post-stress induction). Studies featuring additional interventions unrelated to acute stress induction (such as pharmacological treatments or neuro-stimulation), dual-task paradigms, studies reporting post-hoc group allocation (e.g., classification of participants as cortisol responders vs. non-responders). Participants with comorbid psychiatric disorders alongside depression, anxiety or BPD were permitted. However, participants with physical or psychiatric disorders not comorbid with depression, anxiety or BPD were excluded. Of the 1855 articles identified, 215 articles were selected for full-text review. However, following full-text review only four studies of BPD participants, and no studies of MDD or anxiety, reported requisite data for inclusion in a meta-analysis. Therefore, a meta-analysis was deemed inappropriate to address the main research question, and a systematic review was conducted.

2.2. Systematic review

2.2.1 Search and Retrieval. A second search was conducted (February 4th, 2023) to identify candidate studies for a targeted systematic review. [Shields et al. \(2016\)](#) original search string was amended to identify studies involving participants diagnosed with MDD, GAD, or BPD. This included studies in which symptoms of depression or anxiety were measured, or those involving closely related constructs such as remitted depression or trait anxiety (amendments in italics):

“Trier Social Stress Test” OR “TSST” OR “cold pressor” OR “cold-pressor” OR “acute stress” OR “stress was induced” OR “stress induction” OR “stress manipulation”) AND (“executive function” OR “executive control” OR “cognitive control” OR “response inhibition” OR “cognitive inhibition” OR “selective attention” OR “executive attention” OR “interference control” OR “emotional interference” OR “sustained attention” OR “working memory” OR “set-shifting” OR “task-switching” OR “cognitive flexibility” OR “n-back” OR “OSPAN” OR “AOSPAN” OR “digit span” OR “Sternberg item recognition” OR “color wheel” OR “colour wheel” OR “change detection task” OR “go/no-go” OR “go no-go” OR “stop signal task” OR “Stroop task” OR “Wisconsin Card Sorting Test” OR “trail making test” OR “letter-number sequencing” OR “d2 test of attention”) AND (“Depress*” OR “MDD” OR “anxiety” OR “anxious” OR “borderline” OR “BPD” OR “emotionally unstable personality” OR “EUPD”).

Peer-reviewed studies were identified via English-language keyword searches on the Web of Science, Scopus, PsycINFO, ProQuest and PubMed databases, for papers published between 1st January 1900 and December 31st, 2022. De-duplication was carried out in the Syras ([Scipilot, 2023](#)) screening platform, using a matching algorithm based on title, author and document metadata to identify candidate papers for manual exclusion.

2.2.2 Eligibility Criteria. Following de-duplication, abstracts were dual-screened by the first author and an independent reviewer (L.B.) to identify experimental studies, or reviews thereof, meeting the following criteria: (1) induction of stress in (2) human participants, (3) ≥ 18 years of age, and (4) subsequent completion of a task established to index WM, cognitive flexibility or inhibition, not more than one hour post-stress induction (stress may exert time-dependent effects on EF; [Shields](#)

et al., 2016; Shields 2020). In addition, eligible studies featured participants diagnosed with MDD, GAD or BPD, or included measurements of depression or anxiety symptoms among non-clinical participants using an established measurement method. In keeping with the HiTOP model, only distress disorders (i.e., MDD, GAD and BPD) were

considered for review, although comorbidity with non-distress disorders was permitted. Due to the inclusion of depressive symptoms (i.e., presentations below threshold for MDD), dysthymia was not directly considered in this review. PTSD was also excluded due to its distinctive onset following exposure to acute traumatic stress—implying a unique

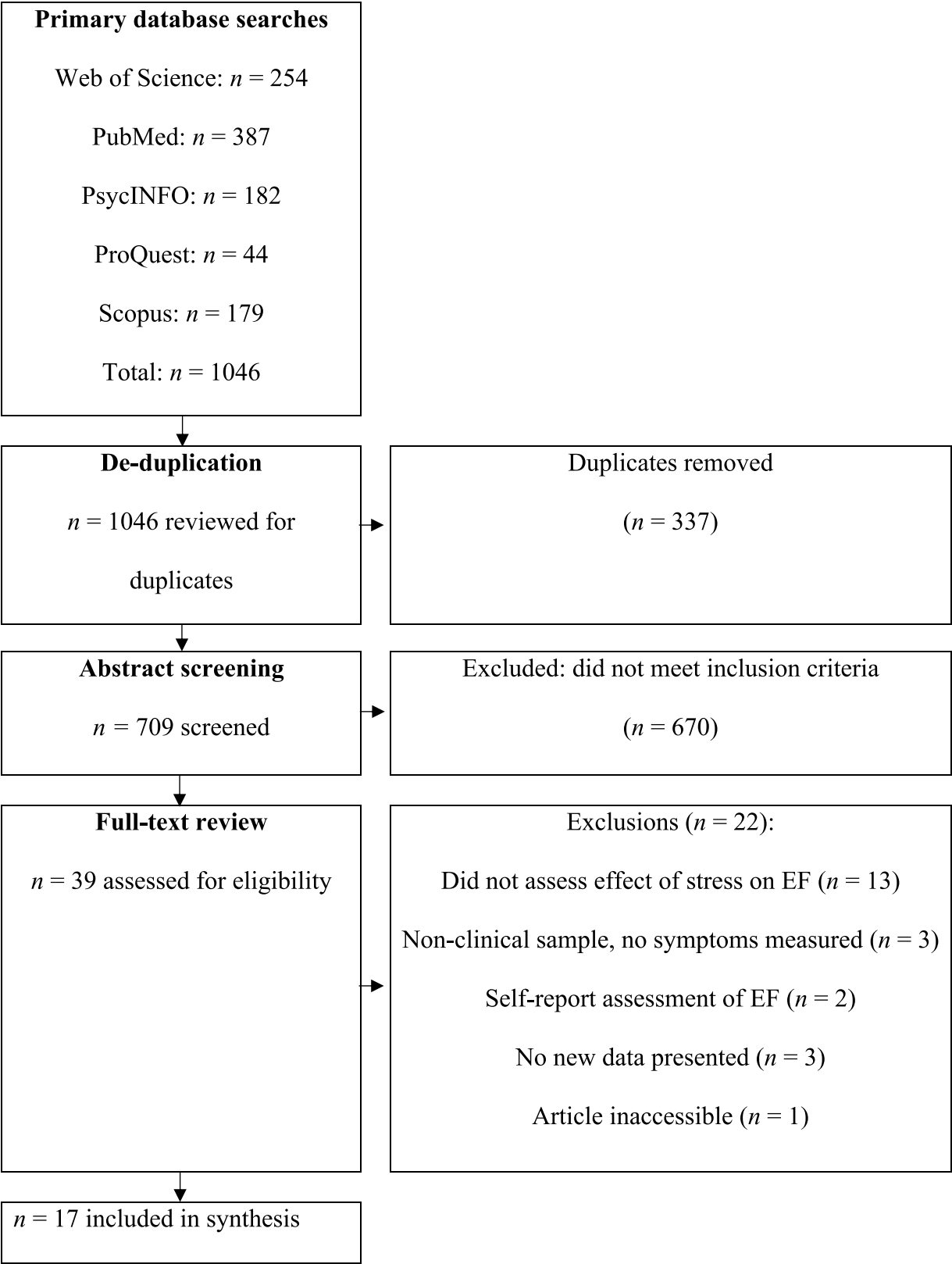


Fig. 1. PRISMA Flow Diagram of Database Search and Study Selection Results.

illness course relative to other distress disorders—as well as recent evidence for its preferential inclusion in a separate ‘traumatic stress’ subfactor (Hyland et al., 2022; Haahr-Pedersen et al., 2023). Studies featuring additional interventions unrelated to acute stress induction (such as antidepressant medication or neurostimulation) were excluded.

The article retrieval, screening and selection process is detailed in Fig. 1 below. Of the 39 studies not excluded during abstract screening, 22 were excluded following full-text review: thirteen did not meet eligibility criteria, three (Aylward et al., 2017; Banks et al., 2014; Vrshek-Schallhorn et al., 2019) reported neither symptom measurement nor clinical diagnosis, and two (Gabrys et al., 2018; Kertz et al., 2017) assessed EF by self-report only. Three narrative reviews (McHugh and Balaratnasingam, 2017; Southwick et al., 2005; Turner et al., 2017) reported no additional statistical analyses. One experimental study identified in the database search (Lange et al., 2012) could not be retrieved, and the authors did not respond to a request for full-text access. All relevant studies cited in these reviews had also been identified in the database searches for the current review.

Among the 17 studies selected for inclusion, five included participants meeting diagnostic criteria for MDD ($n = 2$) or BPD ($n = 5$); two included participants with a prior diagnosis of MDD (remitted MDD), and seven included measurement of known symptoms of depression ($n = 5$) or anxiety ($n = 2$) among non-clinical participants. One study included non-clinical participants with genetic vulnerability to depression, having one or more biological parents with a history of MDD. All included studies were published between 2006 and 2020.

2.2.3 Quality Assessment. Quality assessment was undertaken using the Quality Assessment with Diverse Studies tool (QuADS; Harrison et al., 2021; Appendix B). The QuADS tool permits quality ratings from 0–3 across 13 domains (e.g., clear statement of the research aims; provision of recruitment data), yielding summary quality scores between 0 (low quality) and 39 (high quality). Quality assessments were undertaken by the first author and validated by an independent reviewer (L.B.) for nine randomly selected studies, 52.94 % of those studies included. Inter-rater agreement across the selected studies was 93.17 %, with eight conflicting scores across 117 items resolved through discussion and mutual agreement. Quality assessment results are reported in Table 1. Ratings for individual studies ranged from 19 to 33 ($M = 25.94$, $SD = 4.16$) of a maximum score of 39, reflecting relatively high study quality overall.

2.2.4 Data Extraction. The following data were extracted for each study: sample characteristics, region, recruitment method, selection criteria, number of participants recruited and excluded, age, gender proportion, study design, method of group assignment (where applicable), stress induction and control tasks, EF assessment tasks,

verification of stress induction, method of assessing symptoms or diagnostic status, time delay between stressor administration and EF assessment, additional tasks or interventions, details of participant follow-up, descriptive statistics, test statistics, confidence intervals, p -values and effect sizes. All available data relating to the administration of EF tasks were collected. Descriptive data and study characteristics are presented in Section 3.1. Data extraction was completed by the first author, and all extracted values were verified by an independent reviewer (L.B.) for the nine studies selected for quality assessment validation.

3. Results

Based on prior literature indicating differential effects of stress on WM, inhibition and cognitive flexibility (Shields et al., 2016), and on the presumption of a common basis of stress sensitivity among distress disorders (Clark and Watson, 2006; Kotov et al., 2017; Watson et al., 2008), studies were grouped for synthesis by executive function. In one study (Hood et al., 2015), WM was ostensibly assessed using a Letter-Number Sequencing task. For the present synthesis, this task was re-classified based on several analyses indicating a preferential loading of this task onto a cognitive flexibility, rather than WM factor (Fleming et al., 2016; Himi et al., 2019, 2021; Klauer et al., 2010; Miyake et al., 2000). In total, WM was examined in nine studies, response inhibition in five studies, cognitive inhibition in three studies and cognitive flexibility in three studies. Two studies (Graver and White, 2007; Niemegeers et al., 2019) examined more than one EF domain and were hence included in syntheses of each.

3.1. Participant and study characteristics

A total of 994 participants were included across the studies, with studies ranging from 29–103 participants. Eight studies reported recruitment statistics, and across these the average dropout rate was 12.95 %. The average participant age was 26.53 ($SD = 8.45$) years, and 82.40 % were female. Study characteristics are reported in Table 2.

3.2. Synthesis results

Key findings from the included studies are summarised in Table 3 below for WM, inhibition and cognitive flexibility. Where available, effect size magnitudes are described according to Cohen’s (1988, 1992) guidelines for η_p^2 and β , corrections to Cohen’s guidelines for r suggested by Gignac and Szodorai (2016), and corrections to Cohen’s guidelines for d suggested by Lovakov and Agadullina (2021). For study #15, η_p^2 was derived from the reported F and df statistics based on Fritz et al. (2012) method.

The stress induction tasks utilised in the included studies are reported in Table 4. For each task, individual stressor components are detailed. The majority of studies (76.47 %) featured stress-induction protocols combining a cognitive task with an aversive experience, such threat of negative social evaluation (social-evaluative threat; Dickerson and Kemeny, 2004). For instance, in the Trier Social Stress Test (TSST; Kirschbaum et al., 1993) participants were instructed to deliver an unscribed speech and complete verbal arithmetic tasks while observed by an unresponsive audience, and while ostensibly being filmed (e.g., Duan et al., 2019; Quinn and Joormann, 2020). Similarly, the Maastricht Acute Stress Test (MAST; Shilton et al., 2017) featured a socially-evaluated arithmetic task coupled with pain induction (e.g., Whitton et al., 2017), and the Mannheim Multicomponent Stress Test (MMST; Kolotlyova et al., 2010) featured a cognitive task coupled with exposure to aversive images and sounds and monetary loss (e.g., Krause-Utz et al., 2016). Other methods included thermally induced pain, threat of electric shock, or administration of hormonal or inflammatory stressors (Table 4).

3.2.1 Working Memory. Nine studies (#4, #5, #6, #10, #11, #12,

Table 1

Average Quality Ratings by Domain (range 0–3) and by Total Score (range 0–39), Based on the Quality Assessment with Diverse Studies Tool.

Quality assessment domain	Overall score M (SD)
Theoretical or conceptual underpinning to the research.	3.00 (0.00)
Statement of research aim/s.	2.53 (0.64)
Clear description of research setting and target population.	2.00 (0.99)
The study design is appropriate to address the stated research aim/s.	2.41 (0.51)
Appropriate sampling to address the research aim/s.	1.24 (0.59)
Rationale for choice of data collection tool/s.	1.94 (1.03)
The format and content of data collection tool is appropriate to address the stated research aim/s.	3.00 (0.00)
Description of data collection procedure.	2.94 (0.26)
Recruitment data provided.	0.71 (1.08)
Justification for analytic method selected.	0.65 (1.10)
The method of analysis was appropriate to answer the research aim/s.	2.94 (0.26)
Evidence that the research stakeholders have been considered in research design or conduct.	0.65 (0.51)
Strengths and limitations critically discussed.	1.94 (0.96)

Table 2
Study characteristics.

#	Study	Region	N	Participants	Exclusions (N)	Female %	QuADS score	Study design and comparison groups	EF assessed
1	Cackowski et al. (2014).	Germany	65	Residential psychiatric patients, community volunteers.	Dropout (3); alcohol abuse (1).	100 %	26	Mixed-design: BPD vs. HC.	Response inhibition
2	Carvalho Fernando et al. (2013).	Germany	64	Residential psychiatric patients, community volunteers.	—	100 %	27	Placebo-controlled crossover: BPD vs. HC.	Response inhibition
3	Duan et al. (2019).	China	52	University students.	—	—	20	Mixed-design: HTA vs. LTA.	Cognitive inhibition
4	Duesenberg et al. (2019).	Germany	98	Psychiatric outpatients, community volunteers.	—	100 %	25	Mixed-design: BPD vs. HC.	WM
5	Gaillard et al. (2020).	Switzerland	32	University students, community volunteers.	—	75 %	27	Mixed-design: HV vs. HC.	WM
6	Graver & White (2007).	United States	33	University students.	—	82 % (est.)	24	Mixed-design: SP vs. SP + MDD vs. HC.	WM Cognitive flexibility
7	Hood et al. (2015).	United States	103	University students.	Task non-completion (5); administration error (4).	51 %	28	Mixed-design: Stressor vs. control.	Cognitive flexibility
8	Krause-Utz et al. (2016).	Germany	88	Psychiatric outpatients, community volunteers.	Dropout (1); met exclusion criteria (3); data loss (3).	100 %	27	Mixed-design: BPD vs. ADHD vs. HC.	Response inhibition
9	Krause-Utz et al. (2013).	Germany	60	Psychiatric outpatients, community volunteers.	—	100 %	23	Mixed-design: BPD vs. ADHD vs. BPD + ADHD vs. HC.	Response inhibition
10	Niemegeers et al. (2019).	Belgium	39	Not reported.	—	100 %	22	Placebo-controlled crossover: rMDD vs. HC.	WM Response inhibition Cognitive inhibition
11	Quinn & Joormann (2020).	United States	66	Respondents to university campus advertisements.	Dropout (15), task non-completion (1), data loss (1), outliers (7).	67.1 %	33	Mixed-design: Stressor vs. control.	WM
12	Quinn & Joormann (2015b).	United States	43	University students.	Dropout (8).	62.8 %	31	Within-subjects: Healthy participants.	WM
13	Quinn & Joormann (2015a).	United States	92	University students.	Data loss (8).	67.4 %	31	Within-subjects: Healthy participants.	WM
14	Quinn et al. (2020).	United States	32	Respondents to university campus advertisements.	Task noncompletion (1); met exclusion criteria (2); outlier (4).	69.2 %	32	Mixed-design: Stressor vs. control.	WM
15	Rydmark et al. (2006).	Sweden	29	Female human-services workers.	Dropout (2); met exclusion criteria (11); aberrant neuroimaging findings (2).	100 %	24	Between-groups: MDD vs HC.	WM
16	Tsumura & Shimada (2012).	Japan	35	University students	Outlier (1).	75 %	22	Mixed-design: Cortisol responders vs. non-responders.	Cognitive flexibility
17	Whitton et al. (2017).	United States	63	Not reported.	Met exclusion criteria (1).	68.8 % (est.).	19	Mixed-design: rMDD vs HC.	Cognitive inhibition

Note: ADHD: attention-deficit/hyperactivity disorder, BPD: borderline personality disorder, EF: executive function, HC: healthy controls, HTA: high trait anxiety, HV: high genetic depression vulnerability, LTA: low trait anxiety, MDD: major depressive disorder, rMDD: remitted MDD, SP: social phobia.

#13, #14 and #15) examined WM. All but one confirmed induction of stress by cortisol assay; study #5 reported no validation attempt. Two studies examined MDD patients, one with comorbid social phobia (#6) and one with no reported comorbidities (#15). Following stress induction, WM performance was lower among (non-comorbid) MDD patients relative to controls (moderate effect size; #15). Conversely, no group differences in WM performance following stress induction were observed between the co-morbid group (MDD and social phobia) and healthy controls (#6). Among this group, however, a negative association of small magnitude was reported between depressive symptoms and WM performance, following stress induction (#6). Four separate studies, each from a single research group (#11, #12, #13, #14), examined depressive symptoms among non-clinical participants. Across these studies, depressive symptoms were found to predict decreased WM performance following stress induction. This performance decrease, in turn, predicted a heightening of depressive symptoms during follow-up surveys planned to coincide with examination stress among the sample of university students. However, the magnitudes of these effects could not be determined from the reported data. One study (#4) that examined BPD patients found poorer WM performance among patients relative to healthy controls (small magnitude), independently of stress

induction. Two further studies (#5, #10) examined WM among patients without current distress symptoms: one included participants with an assumed genetic vulnerability to MDD (based on MDD among biological parents; #5) and one including patients with remitted MDD (rMDD; #10). Neither study confirmed group differences in WM performance following a psychosocial stress induction relative to healthy controls. Interestingly, rMDD patients evidenced better WM performance than healthy controls following inflammatory stress induction via *Salmonella typhi* vaccine administration, however the magnitude of this effect could not be inferred from the available data. This finding stands in contrast to one study, reported above, which found post-stress increases in inflammatory cytokines predictive of both WM impairment and depressive symptoms among healthy participants (#14). However, as with (#10), the magnitude of effect could not be inferred from the reported data. Of the included studies, none assessed WM among participants with an anxiety disorder or in relation to anxiety symptoms. Taken together, the available literature indicates that the presence of depressive symptoms is associated with poorer WM performance following acute stress exposure, even for sub-clinical presentations wherein diagnostic criteria for a depressive disorder are not met. It is further indicated that WM performance is poorer among BPD patients relative to healthy controls,

Table 3

Summary of Findings of the Included Studies.

#	Comparison groups	Stressor and control tasks	EF assessed	EF task	Results
1	BPD vs. HC.	MMST. Nil control.	Response inhibition	Go-stop	Compared to HC, BPD patients evidenced higher rates of inhibition failure under stress conditions, $F_{(1,58)} = 6.28, p = .015, \eta_p^2 = 0.10$, but not at rest.
2	BPD vs. HC.	Hydro-cortisone (10 mg) vs. placebo.	Response inhibition	Go/No-Go	Across participants, faster RT was observed for the hydrocortisone group relative to the control group, $F_{(1,62)} = 4.22, p = .044, \eta_p^2 = 0.06$. No differences in RT emerged between BPD and HC, $F_{(1,62)} = 0.22, p = .64$. No effects were observed on false alarm rates.
3	HTA vs. LTA.	TSST. Nil control.	Cognitive inhibition	Flanker	A group-by-time interaction was observed for RT interference, $F_{(1,50)} = 20.38, p < .001, \eta_p^2 = 0.29$, with HTA participants responding slower than LTA participants prior to stress induction ($p = .01$), and faster after ($p = .002$). No group differences in error rate were observed.
4	BPD vs. HC.	TSST vs. TSST-P.	WM	Word Suppression Test	No effect of stress on WM was observed. However, the BPD group evidenced poorer WM performance than HCs for both neutral and negatively-valenced stimuli, $F_{(1,90)} = 5.59, p = .02, \eta_p^2 = 0.06$, and $F_{(1,90)} = 4.21, p = .04, \eta_p^2 = 0.05$, respectively.
5	HV vs. HC.	Shock-threat vs. no-threat conditions.	WM	Fribourg Reward Task	No group differences between depression-vulnerable and HC participants were observed, and the inclusion of depression scores as a covariate yielded no significant effect.
6	SP vs. SP + MDD vs. HC.	TSST. Nil control.	WM Cognitive flexibility	Digit Span, Spatial Span TMT-B, WCST	Across groups, participants in the shock-threat condition exhibited greater accuracy, $F_{(1,30)} = 8.00, p \leq 0.01, \eta_p^2 = 0.21$, and faster RT, $F_{(1,30)} = 17.90, p \leq 0.001, \eta_p^2 = 0.37$ relative to the no-shock condition. WM: No group effects were observed for digit span, despite an overall pre- to post-stressor improvement, $F_{(1,30)} = 5.97, p < .05, \eta_p^2 = 0.17$. In the stress condition, spatial span was improved among HC but impaired among SP patients, $F_{(2,30)} = 3.52, p < .05, \eta_p^2 = 0.19$. Flexibility: Across groups, TMT-B completion time was faster after stress induction than before, $F_{(1,30)} = 31.11, p < .001, \eta_p^2 = 0.51$, and this effect was greater for the HC and comorbid groups than for the SP group, $F_{(2,30)} = 4.09, p < .05, \eta_p^2 = 0.21$. Participants in the comorbid group, despite evidencing the greatest reduction in completion time, committed more errors than the SP and HC groups, $\chi^2(2, N = 33) = 11.79, p < .01$. In the WCST, stress lead to decreased errors among the comorbid and HC groups, $F_{(2, 28)} = 4.23, p < .05, \eta_p^2 = 0.23$ and, across groups, an overall decrease in error rate, $F_{(1,28)} = 4.45, p < .05, \eta_p^2 = 0.14$. In the stress condition, BDI scores were positively associated with TMT-B errors, $r = 0.37, p < .05, n = 33$, and negatively associated with the spatial span scores, $r = -0.39, p < .01, n = 33$.
7	HC: Stress vs. control.	Cold-pressor task vs. warm water task.	Flexibility	LNS	No main effect of stress condition was observed. In the stress condition, LNS scores and anxiety were negatively associated during stress, $r = -.31, p = .03$, and after a 15-minute rest period, $r = -.34, p = .01$. A moderated mediation analysis indicated that anxiety mediated this relationship among participants evidencing high cortisol responses.
8	BPD vs. ADHD vs. HC.	MMST. Nil control.	Response inhibition	Go-stop, IMT	IMT scores suggested poorer inhibition performance among both BPD and ADHD groups relative to HC, $F_{(2,85)} = 7.63, p = .001, \eta_p^2 = 0.15$. Among BPD participants only, inhibition was further impaired following stress induction, $t_{(29)} = 2.08, p = .046, d = 0.36$. No effects pertaining to BPD were evident on the Go-stop task.
9	BPD vs. ADHD vs. BPD + ADHD vs. HC.	PASAT. Nil control.	Response inhibition	IMT, DMT	IMT scores revealed poorer inhibition among the comorbid (BPD + ADHD) and ADHD groups, but not BPD, relative to HC, $F_{(3,56)} = 6.57, p = .001, \eta_p^2 = 0.26$. However, stress induction lead to enhanced inhibition among the BPD group, $t_{(14)} = 2.90, p = .013, d = 0.91$. No effects were observed for DMT ratio.
10	rMDD vs. HC.	TSST vs. TSST-P; <i>S. typhi</i> vaccine (inflammatory stressor) vs. placebo.	WM Response inhibition Cognitive inhibition	Digit Span Backward CPT Emotional Stroop	WM: Among rMDD patients, but not HC, inflammatory challenge via vaccine administration led to an increase in digit span, $B = 1.21, 95\%CI = 1.96, p = .016$, and a slowing of CPT reaction time, $B = 25.3, 95\%CI = 21.8, p = .026$. No effect was observed for CPT accuracy. No effect of TSST was observed. Inhibition: Among rMDD patients only, vaccine administration lead to an increase in completion time for negative trials, $B = 5.51, 95\%CI = 4.71, p = .025$, but not in positive or neutral trials. No effect of TSST was observed.
11	HC: Stress vs. control.	TSST vs. TSST-P.	WM	N-back	No effect of stress on n-back performance was observed. However, decreased n-back performance under stress was found to predict the later emergence of depressive symptoms (BDI-II) under contextual stress (university exams), $B = 0.76, t_{(45)} = 2.33, p = .025$.
12	HC	TSST. Nil control.	WM	N-back	No effect of stress on n-back performance was observed. Depression symptoms (BDI-II) predicted post-stress error rate at the trend level, $B = 0.14, \beta = 0.33, t_{(42)} = 1.86, p = .07$, an effect not evident for pre-stress error rates, $B = -0.02, \beta = -0.04, t_{(42)} = -0.22, p = .82$. Decreased n-back performance under stress was associated with an increase in depression symptoms at follow-up, $\Delta R^2 = 0.10, \Delta F_{(1,38)} = 7.89, p < .01$. An increase in n-back errors was observed from pre- to post-stress, $t_{(91)} = 2.62, p = .01, d = 0.28$. Depression symptoms (BDI-II) were more
13	HC	TSST. Nil control.	WM	N-back	

(continued on next page)

Table 3 (continued)

#	Comparison groups	Stressor and control tasks	EF assessed	EF task	Results
14	HC	TSST vs. TSST-P.	WM	N-back	strongly related to error rates at post-stress than pre-stress, $z_{(89)} = 2.76$, $p < .01$. Post-stress errors were associated with depression symptoms at high levels of ruminative brooding, $\beta = 0.17$, $t_{(86)} = 3.34$, $p = .001$, but not at low levels, $\beta = -0.03$, $t_{(86)} = -0.48$, $p = .63$. No effect of stress on n-back performance was assessed. Task errors at post-stress were associated with salivary interleukin-6 cytokine (IL-6) levels, $B = 1.15$, $\beta = 0.74$, $t_{(31)} = 2.27$, $p = .044$. In the stress condition only, greater increases in IL-6 were associated with higher depressive symptoms, $B = 1.32$, $\beta = 0.92$, $t_{(31)} = 2.60$, $p = .014$.
15	MDD vs. HC.	Dexamethasone (1.5 mg) and CRH (0.1 mg). Nil control.	WM	Digit Span Backwards	WM: Following CRH administration, digit span was poorer among depressed patients relative to HCs, $F_{(1,52)} = 11.6$; $p = .001$.
16	HC: Cortisol responders vs. non-responders.	Mental arithmetic task. Nil control.	Cognitive flexibility	TMT-B	No TMT-B performance differences were observed between cortisol responders and non-responders following stress induction. No association with depression symptoms was observed.
17	rMDD vs HC.	MAST. Nil control.	Cognitive inhibition	Flanker	Across groups, RTs were decreased at post-stress vs. pre-stress, $F_{(1,61)} = 29.96$, $p < .001$, $\eta_p^2 = 0.33$, and on congruent vs. incongruent trials, $F_{(1,61)} = 438.73$, $p < .001$, $\eta_p^2 = 0.88$. No group differences reported.

Note: ADHD: attention-deficit/hyperactivity disorder, BDI: Beck Depression Inventory (second edition), BPD: borderline personality disorder, CPT: Continuous Performance Task, CRH: corticotropin releasing hormone, DMT: Delayed Memory Task, HC: healthy controls, HTA: high trait anxiety, HV: high genetic depression vulnerability, IMT: Immediate Memory Task, LNS: Letter-Number Sequencing, LTA: low trait anxiety, MAST: Maastricht Acute Stress Test, MDD: major depressive disorder, MMST: Mannheim Multicomponent Stress Test, PASAT: Paced Auditory Serial Addition Task, rMDD: remitted MDD, RT: reaction time, SP: social phobia, TMT-B: Trail-Making Test (part B), TSST: Trier Social Stress Test, TSST-P: TSST placebo task (non-stressful), WCST: Wisconsin Card Sorting Task, WM: working memory.

Table 4
Included Stress Induction Tasks Classified by Individual Stressor Components.

#	Stress induction task	Applied pain	Aversive images	Aversive noise	Cognitive task	Social evaluation	Pharmacological induction	Threat of pain
7	Cold-pressor task	×						
15	Dexamethasone & CRH administration						×	
2	Hydrocortisone administration						×	
17	MAST	×			×	×		
16	Mental arithmetic				×	×		
1, 8	MMST		×	×	×			
9	PASAT			×	×			
5	Shock-threat task							×
3, 4, 6, 10, 11, 12, 13, 14	TSST				×	×		
10	Vaccine administration						×	

Note: CRH: corticotropin releasing factor, MAST: Maastricht Acute Stress Test, MMST: Mannheim Multicomponent Stress Test, PASAT: Paced Auditory Serial Addition Task, TSST: Trier Social Stress Test.

regardless of acute stress induction.

3.2.2 Inhibition. Five studies (#1, #2, #8, #9 and #10) examined response inhibition, and three studies (#3, #10 and #17) examined cognitive inhibition. Four studies confirmed stress induction using cortisol assay, two (#1 and #8) using a combination of heart rate measurement and self-report, and one (#9) using self-report only. Four studies (#1, #2, #8, #9) examined response inhibition among BPD patients compared to healthy controls, and in relation to attention-deficit/hyperactivity disorder (#1, #8, #9) and posttraumatic stress disorder (#2). Findings were mixed regarding the differential effects of stress induction on response inhibition. Two studies (#1, #8) reported poorer response inhibition among BPD patients relative to controls (small magnitude), following stress induction. Another study (#9) reported better response inhibition among patients compared to controls (large magnitude), following stress induction, while another (#2) reported no differential effect of stress induction on inhibition. One study (#10) examined the effects of both psychosocial and inflammatory (*S. typhi* vaccine) stressors on response and cognitive inhibition among rMDD patients. While no effect of psychosocial stress induction was reported, poorer response inhibition and greater cognitive inhibition was observed among rMDD patients relative to controls (no effect sizes reported). However, another study comparing rMDD patients with

controls (#17) found no group effect of stress induction on cognitive inhibition. Finally, one study (#3) compared the effects of acute stress on cognitive inhibition between high and low trait anxiety participants. This study reported a group-by-time interaction of large magnitude, with cognitive inhibition impaired at pre-stress but improved at post-stress, among high relative to low trait anxiety participants.

To summarise, findings were mixed and suggested that both response and cognitive inhibition are differentially affected by acute stress in distress disorders relative to controls. Further research is needed to clarify the nature and direction of these effects.

3.2.3 Cognitive Flexibility. Three studies (#6, #7 and #16) examined cognitive flexibility. One study compared healthy controls and social phobia patients with, and without, comorbid MDD (#6). The other studies examined depressive (#16) and anxious (#7) symptoms among non-clinical participants. All three confirmed stress induction via cortisol assay. No differential effect of acute stress induction was reported between MDD and non-MDD participants (#6). However, independently of stress induction, MDD patients evidenced faster and more error-prone task completion than non-MDD participants (no effect size reported). Furthermore, a positive association between depression scores and error rates was reported (small magnitude) (#6). Conversely, no effect of stress induction, nor association between depression

symptoms and cognitive flexibility performance, was observed in study #16. Among healthy participants, a negative association of small magnitude was observed between anxiety symptoms and cognitive flexibility performance under stress. This effect was found to be moderated by stress-related increases in salivary cortisol (#7).

In sum, a lack of available data precludes even tentative conclusions regarding the hypothesised interplay between distress disorders, or symptoms, and cognitive flexibility performance under stress.

4. Discussion

This systematic review examined the literature on acute stress-induced alterations in executive functions (EF) in the context of distress disorders, namely MDD, GAD, BPD, and their related symptoms. The primary aim was to determine whether differential patterns of EF alterations are observed in these distress, and in the presence of associated symptoms, compared to healthy and non-clinical populations. We examined studies comparing the effects of acute stress induction on EF task performance between MDD, GAD and BPD patients and healthy controls. In keeping with emerging dimensional approaches to psychopathology we also examined associations between stress-induced EF alterations and symptoms of depression and anxiety amongst non-clinical participants. Seventeen studies were included in separate syntheses of WM, cognitive inhibition, response inhibition and cognitive flexibility.

4.1. Working memory

Our results suggest that depression may be associated with heightened susceptibility to WM disruption under stress. Overall, both MDD and elevated depression symptoms among non-clinical participants were found to be moderately associated with stress-induced impairment of WM. Furthermore, among non-clinical participants, this impairment predicted the later emergence of depressive symptoms during an extended period of heightened stress (in this case, examination stress among students). Conversely, neither remitted MDD nor genetic vulnerability to MDD were associated with WM alterations. The apparent absence of stress-related WM impairment among these non-symptomatic groups suggests that susceptibility to impairment emerges coextensively with the onset of depressive symptoms. EF impairments under non-stressful conditions are widely reported in both active and remitted MDD (Nuño et al., 2021; Snyder, 2013), and have been theorised to play a causal role in the expression of MDD symptoms (Snyder, 2013). The findings of the current synthesis complement this perspective, suggesting that trait neurocognitive vulnerabilities contributing to symptom expression may be compounded by heightened susceptibility to stress-induced EF impairment during active depression.

None of the included studies examined WM in the context of anxiety. Only one (#4) examined WM among BPD patients, reporting a small effect of impaired WM among patients regardless of stress induction. However, control participants in this study underwent a placebo task known to induce mild stress responses even among healthy participants (Het et al., 2009). Given the heightened stress reactivity typical of BPD patients (Stepp et al., 2016), it could be that group differences in stress-induced WM alterations were masked by BPD patients' reactivity to the control task. To summarise, depressive symptoms were associated with increased susceptibility to WM impairment under stress; however, conclusions regarding anxiety and BPD await further research.

4.2. Inhibition

Support for group differences in stress-induced inhibition alterations was inconsistent. According to the framework proposed by Shields et al. (2016), induction of acute stress is associated with improved response inhibition and impaired cognitive inhibition. Neither of the two studies examining cognitive inhibition among remitted MDD patients supported

such an effect. One study comparing non-clinical participants with high- and low-trait anxiety (#3) reported pre-stress impairment and post-stress improvement of cognitive inhibition among high- relative to low-trait anxiety participants, utilising a Flanker task. However, this effect pertained to reaction time only, and no group difference was observed for task accuracy. We contend, therefore, that the reported effect pertains to response rather than cognitive inhibition.

Response inhibition was most widely examined among BPD patients. Two studies (#1, #8) reported modest impairment of response inhibition under stress. Conversely, one study (#9) found that stress induction improved response inhibition, with a large effect size, among BPD patients relative to healthy controls. In the latter study, moderate stress was induced by verbal completion of a neurocognitive assessment (Diehr et al., 1998). In contrast, studies #1 and #8 utilised the Mannheim Multicomponent Stress Test, a comprehensive stress induction protocol combining cognitive, affective, acoustic and motivational stressors (Reinhardt et al., 2012). Therefore, these seemingly opposite findings can be reconciled in terms of the purported adaptation of EFs to support cognition under moderate stress, and to prioritise action under severe stress (Arnsten, 2009; Yuen et al., 2009; Sapolsky, 2015; Oshri et al., 2022). Among healthy participants, response inhibition is improved under severe stress (Shields et al., 2016). Conversely, among BPD patients, response inhibition was improved under relatively moderate stress (#9) but impaired under more severe stress (#1, #8). This is in keeping with the broader phenomenology of BPD, in which heightened stress reactivity (Stepp et al., 2016) and stress-linked behavioural disinhibition (Nigg et al., 2005) are core features. A similar effect could account for the reversal of relative inhibitory control between high- and low-trait anxiety participants following stress induction (#3). However, further research is needed to qualify this suggestion, and to better understand how inhibitory control interacts with acute stress within the broader spectra of distress disorders.

4.3. Cognitive flexibility

The synthesis of cognitive flexibility studies was inconclusive. No direct evidence that acute stress differentially affects cognitive flexibility performance in distress disorder patients relative to controls was found. However, modest associations between depressive (#6) and anxious (#7) symptoms and impaired cognitive flexibility were observed among healthy participants. Intriguingly, the association between anxiety and cognitive flexibility impairment was mediated by salivary cortisol levels, suggesting that distress pathology may indeed be associated with impaired cognitive flexibility. Conversely, a comparison of healthy cortisol responders and non-responders (#16) yielded no performance differences and no associations with depression symptoms, although such an effect may have been confounded by the aberrant cortisol reactivity elsewhere observed in MDD (Nandam et al., 2020). Accordingly, no general conclusions could be drawn regarding the presence, or otherwise, of altered cognitive flexibility under stress in the context of distress disorders.

4.4. Theoretical implications

Our findings suggest an increased susceptibility to stress-induced EF alterations (Shields et al., 2016) in the presence of distress disorder symptoms. The strongest support is from studies of WM in depression. However, a lack of available studies precluded generalization of these findings to other distress disorders (i.e., GAD and BPD), or to cognitive inhibition and cognitive flexibility. Further to predictions, the current review indicated impaired response inhibition under stress among BPD patients compared to controls. Collectively, the evidence for a common susceptibility to EF disruption under acute stress in distress disorders remains tentative.

Susceptibility to WM disruption was observed among both MDD patients and non-clinical participants with depressive symptoms. This is

in keeping with a core premise of the HiTOP model, which describes symptom severity as reflecting a degree of dysfunction along a continuum from healthy to disordered regardless of presumed diagnostic thresholds (Kotov et al., 2017, 2021). Furthermore, this susceptibility was observed in the presence of depression symptoms, but not in their absence (as in remitted MDD or genetic vulnerability). EF deficits under non-stressful conditions have previously been reported in MDD (Nuño et al., 2021; Snyder, 2013), and it has been suggested that trait neurocognitive vulnerabilities interact with environmental contingencies, such as stress, to play a causal role in MDD pathogenesis (Snyder, 2013). However, strict neurobiological models of emotional disorders have been critiqued as implying explanatory priority of biological over psychological or contextual factors (Borsboom et al., 2019). The current review lends emphasis to the role of situational stressors in the pathogenesis and treatment of distress disorders.

4.5. Clinical implications

Our findings suggest a differential effect of acute stress on EF for individuals with distress disorder symptoms relative to those without. Acute stress is thought to enhance certain cognitive processes at moderate levels, and to disrupt them at more severe levels according to an inverted-U function (Yuen et al., 2009; Sapolsky, 2015; Oshri et al., 2022). Under moderate stress, for instance, enhancements in WM and other EFs may confer improvements in complex decision-making, problem-solving and social functioning (McQuade et al., 2013; Otto et al., 2013). Under heightened stress, however, prefrontal functioning is diminished, allowing a relative dominance of subcortical processes which favour rapid execution of reflexive or well-rehearsed behaviours (Arnsten, 2009). This is thought to be adaptive in time-sensitive contexts in which action is preferable to deliberative cognitive engagement. In distress disorders, which are characterised by altered stress reactivity (Clark and Watson, 2006; Kotov et al., 2021), EF adaptation may be atypical. For example, under moderate stress MDD patients may experience WM impairment rather than improvement, and under severe stress BPD patients may experience behavioural disinhibition rather than improved inhibitory control.

Since EFs are crucial to the coordination of emotion regulation strategies such as distraction and reappraisal (Joormann and Gotlib, 2010; Ochsner et al., 2012; Toh and Yang, 2022a), atypical EF adaptation may contribute to impaired emotion regulation in the presence of distress disorder symptoms. Susceptibility to disruption of WM or inhibition under moderate (rather than severe) stress could help explain non-responsiveness to cognitive therapies among many distress disorder patients (Borkovec and Ruscio, 2001; Cristea et al., 2017; Renna et al., 2017; Cuijpers et al., 2013). According to the process model of emotion regulation (Gross, 2015; Sheppes and Gross, 2011), distraction and reappraisal typically operate early in the emotion-generative process, limiting the full expression of the target emotional response. Early timing is critical: as an emotional response develops through time it becomes more intense and requires greater efforts at modulation (Sheppes and Gross, 2011). Susceptibility to impairment of WM and inhibition under stress may translate to faster loss of these capacities as the emotional response to a stressor develops over time. This may disrupt the coordination and execution of emotion regulation strategies in the critical early moments of the emotion-generative process. Susceptibility to such effects could be a substantial treatment-interfering factor, and empirical evaluation of this possibility could help account for the modest effectiveness of cognitive therapies on MDD and other distress disorders.

4.6. Limitations

Among the available studies, heterogeneity in study design, stress induction methods, EF assessment and reporting of methodological details limit the generalisability of the reported effects. Many studies, for

instance, did not control for confounds such as medication use. A further consideration is heterogeneity in stress induction methods across the included studies. Most studies utilised established stress-induction procedures such as the TSST (Kirschbaum et al., 1993), and all but two used physiological measures to verify stress induction: cortisol assay (13 studies), or heart rate changes in conjunction with self-report (2 studies). Cortisol assay is thought to reflect activity of the hypothalamic-pituitary-adrenal axis (HPA; Russell et al., 2012), whereas cardiac changes are thought to reflect autonomic nervous system (ANS) activity (Kim et al., 2018). HPA activation is widely considered a “benchmark” measure of stress induction, though may more specifically reflect social-evaluative threat (Dickerson and Kemeny, 2004; Roos et al., 2017). Twelve of the included studies sought to induce stress through either direct pharmacological modulation of the HPA or protocols featuring social-evaluative stress. Appropriately, all such studies utilised cortisol assay to confirm stress induction. Furthermore, findings from study #7 suggested that anxiety mediated the deleterious effect of acute stress on WM only when increased cortisol was observed (Hood et al., 2015). Among the studies utilising heart rate as a manipulation check, no rationale for this method was provided. Finally, the single study to include no manipulation check (#5) did report reduced activation of the nucleus accumbens under unpredictable shock threat among depression-vulnerable participants relative to controls, under low but not high WM load (Gaillard et al., 2020). Thus, notwithstanding the possibility of publication bias, the available studies were generally successful in confirming stress induction. However, the evidence that different types of stressor stimuli may preferentially elicit stress responses in different physiological systems poses a challenge to the consideration of acute stress as a unitary construct. Specificity with regard to stressor stimuli and their effects on cognition and physiology is encouraged for future research (Allen et al., 2014; Godoy et al., 2018; Shields, 2020). With regard to investigating the role of acute stress in psychopathology, particular attention is needed to ensure that experimental stress manipulation procedures correspond sufficiently to real-world stressors, and that presumed stressors are closely specified (Epel et al., 2018).

Similarly, and as previously observed (Shields et al., 2016; Roos et al., 2017), the timing between the onset of stressor tasks, manipulation checks and EF task administration was highly variable between studies, and in many cases not reported. Stress elicits time-dependent effects across many physiological systems, including the HPA axis, ANS, and systems involved in regulating catecholamines, sex hormones and immune response, each with differing time-courses (Shields, 2020). Precise specification of protocol timing is therefore crucial in establishing the relation between a given stimuli, a cognitive process of interest, and a physiological mechanism of effect. In the present study, EF adaptations were hypothesised to result from a rapid deactivation of prefrontal systems following stress exposure. However, the primacy of this mechanism remains speculative, and the observed effects may be better attributable to other physiological processes. Greater specificity in regards to protocol timing may allow attribution of stress effects on EF to specific mechanisms, and possibly enable identification of errant physiological processes in psychological disorders (for a review of key considerations in experiment design, see Shields, 2020).

Finally, some limitations of the review process must be noted. Neither dysthymia nor PTSD was included amongst the distress disorders examined. However, the inclusion of psychiatric comorbidities presents a possible confounding factor, particularly in light of the high comorbidities amongst distress disorders (Kotov et al., 2021). Gender- and age-dependent effects were not considered in the current synthesis (c.f. Shields et al., 2016). Notably, the high proportion of female to male participants and the relatively restricted age range among the included studies limits the generalisability of the findings, as does the exclusion of studies including children. Similarly, most of the included studies appeared to draw on samples from Western, educated, industrialized, rich, and democratic populations (Heinrich et al., 2010). While the

included quality assessment indicated high overall quality among the sampled studies, the conclusions remain tentative without a more comprehensive assessment of risk of bias. Further, the search terms utilised may not have captured all relevant studies and supplementary means of identifying target articles, such as contacting authors or screening reference lists of the identified studies may have revealed additional studies. Further research is needed to better understand the commonalities and differences between acute stress effects on EFs among MDD, GAD and BPD before firm conclusions can be drawn. Despite the focus in this review on distress psychopathology, alterations in normative stress-linked EF effects are not presumed to be unique to these disorders, and complementary investigations of other conditions, such as the ‘fear disorders’ (panic disorder, social anxiety disorder and specific phobia) may prove fruitful. Similarly, examination of acute stress effects on “hot” EFs such as reward processing and delay discounting may provide further context to the findings reported herein. Heterogeneity of study designs represents a challenge, and interested researchers are encouraged to consider standardisation of stress induction procedures and EF tasks to promote generalisability of findings. Multimethod validation of stress induction may improve reliability, and a standardised protocol to capture the time-course of stress-induced EF alterations could be of particular relevance. Finally, exploration of the roles of age, gender and similar factors is recommended, particularly as gender has previously been found to moderate the effects of acute stress on EF (Shields et al., 2016) and young adults were overrepresented across the included studies.

5. Conclusion

The findings of this review indicate increased susceptibility to stress-induced WM and inhibition disruption in, respectively, depression and

BPD. Interference with executive processes under stress may undermine the efficacy of treatments predicated on emotion regulation strategies, and may be a ubiquitous and under-appreciated feature of distress disorders. More studies are needed to understand the effects of stress on other executive processes, and whether these effects pertain to distress disorders generally.

CRediT authorship contribution statement

T.M. Scott: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joanne M. Dickson:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Checklist (Page et al., 2021)

Section and topic	Item #	Checklist item	Comment
Title			
Title	1	Identify the report as a systematic review.	Completed.
Abstract			
Abstract	2	Follow the PRISMA 2020 for Abstracts checklist (Page et al., 2021).	The abstract was written in accordance with the journal guidelines and specified word count.
Introduction			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Completed.
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Completed.
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Completed.
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Completed.
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Completed.
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Completed.
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Completed.
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in	Completed.

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Section and topic	Item #	Checklist item	Comment
Study risk of bias assessment		each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Completed.
	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Completed.
	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	Completed.
Effect measures			
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Completed.
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Completed.
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Completed.
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model (s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Completed. No meta-analysis was performed due to a lack of studies suitable for meta-analysis.
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression).	The review process indicated a high degree of heterogeneity of study designs and measures. A systematic review was conducted due to a lack of studies suitable for meta-analysis.
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	A systematic review was conducted due to a lack of studies suitable for meta-analysis.
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	A quality assessment tool suitable for diverse study methods was utilised.
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	A quality assessment tool suitable for diverse study methods was utilised.
Results			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Completed. PRISMA flow diagram was included.
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Completed.
Study characteristics	17	Cite each included study and present its characteristics.	Completed.
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Quality assessment scores were reported for each study.
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	Completed.
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Completed.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	For the reasons provided, no meta-analyses were completed. A narrative synthesis of effects was completed in keeping with a systematic review.
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	As addressed above (item 13e).
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Sensitivity analyses were not performed due to a lack of studies suitable for meta-analysis.
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Quality assessment scores were reported for each study, and overall quality scores were reported for each domain of the quality assessment tool.
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Quality assessment scores were reported for each study, and overall quality scores were reported for each domain of the quality assessment tool.
Discussion			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Completed.
	23b	Discuss any limitations of the evidence included in the review.	Completed.
	23c	Discuss any limitations of the review processes used.	Completed.
	23d	Discuss implications of the results for practice, policy, and future research.	Completed.
Other information			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	The authors omitted to pre-register this systematic review.
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	The authors omitted to prepare a review protocol.

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Section and topic	Item #	Checklist item	Comment
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	No amendments to declare.
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Completed.
Competing interests	26	Declare any competing interests of review authors.	Completed.
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	The data extraction table will be made available upon request.

Appendix B

Quality Assessment with Diverse Studies (QuADS) Rating Guidelines (Harrison et al., 2021)

Criteria	Score: 0	Score: 1	Score: 2	Score: 3
1. Theoretical or conceptual underpinning to the research	No mention at all.	General reference to broad theories or concepts that frame the study, e.g., key concepts were identified in the introduction section.	Identification of specific theories or concepts that frame the study and how these informed the work undertaken, e.g., key concepts were identified in the introduction section and applied to the study.	Explicit discussion of the theories or concepts that inform the study, with application of the theory or concept evident through the design, materials and outcomes explored, e.g., key concepts were identified in the introduction section and the application apparent in each element of the study design.
2. Statement of research aim/s	No mention at all.	Reference to what the sought to achieve embedded within the report but no explicit aims statement.	Aims statement made but may only appear in the abstract or be lacking detail.	Explicit and detailed statement of aim/s in the main body of report.
3. Clear description of research setting and target population	No mention at all.	General description of research area but not of the specific research environment, e.g., 'in primary care'.	Description of research setting is made but is lacking detail, e.g., 'in primary care practices in region [x]'.	Specific description of the research setting and target population of study, e.g., 'nurses and doctors from GP practices in [x] part of [x] city in [x] country'.
4. The study design is appropriate to address the stated research aim/s	No research aim/s stated or the design is entirely unsuitable, e.g., a Y/N item survey for a study seeking to undertake exploratory work of lived experiences.	The study design can only address some aspects of the stated research aim/s, e.g., use of focus groups to capture data regarding the frequency and experience of a disease.	The study design can address the stated research aim/s but there is a more suitable alternative that could have been used or used in addition, e.g., addition of a qualitative or quantitative component could strengthen the design.	The study design selected appears to be the most suitable approach to attempt to answer the stated research aim/s.
5. Appropriate sampling to address the research aim/s	No mention of the sampling approach.	Evidence of consideration of the sample required, e.g., the sample characteristics are described and appear appropriate to address the research aim/s.	Evidence of consideration of sample required to address the aim, e.g., the sample characteristics are described with reference to the aim/s.	Detailed evidence of consideration of the sample required to address the research aim/s, e.g., sample size calculation or discussion of an iterative sampling process with reference to the research aims or the case selected for study.
6. Rationale for choice of data collection tool/s	No mention of rationale for data collection tool used.	Very limited explanation for choice of data collection tool/s, e.g., based on availability of tool.	Basic explanation of rationale for choice of data collection tool/s, e.g., based on use in a prior similar study.	Detailed explanation of rationale for choice of data collection tool/s, e.g., relevance to the study aim/s, co-designed with the target population or assessments of tool quality.
7. The format and content of data collection tool is appropriate to address the stated research aim/s	No research aim/s stated and/or data collection tool not detailed.	Structure and/or content of tool/s suitable to address some aspects of the research aim/s or to address the aim/s superficially, e.g., single item response that is very general or an open-response item to capture content which requires probing.	Structure and/or content of tool/s allow for data to be gathered broadly addressing the stated aim/s but could benefit from refinement, e.g., the framing of survey or interview questions are too broad or focused to one element of the research aim/s.	Structure and content of tool/s allow for detailed data to be gathered around all relevant issues required to address the stated research aim/s.
8. Description of data collection procedure	No mention of the data collection procedure.	Basic and brief outline of data collection procedure, e.g., 'using a questionnaire distributed to staff'.	States each stage of data collection procedure but with limited detail or states some stages in detail but omits others, e.g., the recruitment process is mentioned but lacks important details.	Detailed description of each stage of the data collection procedure, including when, where and how data was gathered such that the procedure could be replicated.
9. Recruitment data provided	No mention of recruitment data.	Minimal and basic recruitment data, e.g., number of people invited who agreed to take part.	Some recruitment data but not a complete account, e.g., number of people who were invited and agreed.	Complete data allowing for full picture of recruitment outcomes, e.g., number of people approached, recruited, and

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Criteria	Score: 0	Score: 1	Score: 2	Score: 3
10. Justification for analytic method selected	No mention of the rationale for the analytic method chosen.	Very limited justification for choice of analytic method selected, e.g., previous use by the research team.	Basic justification for choice of analytic method selected, e.g., method used in prior similar research.	who completed with attrition data explained where relevant. Detailed justification for choice of analytic method selected, e.g., relevance to the study aim/s or comment around of the strengths of the method selected.
11. The method of analysis was appropriate to answer the research aim/s	No mention at all.	Method of analysis can only address the research aim/s basically or broadly.	Method of analysis can address the research aim/s but there is a more suitable alternative that could have been used or used in addition to offer a stronger analysis.	Method of analysis selected is the most suitable approach to attempt answer the research aim/s in detail, e.g., for qualitative interpretative phenomenological analysis might be considered preferable for experiences vs. content analysis to elicit frequency of occurrence of events.
12. Evidence that the research stakeholders have been considered in research design or conduct.	No mention at all.	Consideration of some the research stakeholders, e.g., use of pilot study with target sample but no stakeholder involvement in planning stages of study design.	Evidence of stakeholder input informing the research. e.g., use of pilot study with feedback influencing the study design/ conduct or reference to a project reference group established to guide the research.	Substantial consultation with stakeholders identifiable in planning of study design and in preliminary work, e.g., consultation in the conceptualisation of the research, a project advisory group or evidence of stakeholder input informing the work.
13. Strengths and limitations critically discussed	No mention at all.	Very limited mention of strengths and limitations with omissions of many key issues. e.g., one or two strengths/limitations mentioned with limited detail.	Discussion of some of the key strengths and weaknesses of the study but not complete, e.g., several strengths/limitations explored but with notable omissions or lack of depth of explanation.	Thorough discussion of strengths and limitations of all aspects of study including design, methods, data collection tools, sample & analytic approach.

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